Course Project Part 2

Question 1:

Add 3 new APIs and their test cases to support removing nodes and edges

- Remove a node API (5 points): removeNode(String label)
- Remove nodes API (5 points): removeNodes(String[] label)
- Remove an edge API (10 points): removeEdge(String srcLabel, String dstLabel)

```java
// removes a node and cleans edges
public void removeNode(String nodeLabel){
    if(nodeLabel==null||nodeLabel.trim().isEmpty()){
        throw new IllegalArgumentException(s: "bad input");
    }
    String clean=nodeLabel.trim();
    if(!nodes.remove(clean)){
        throw new IllegalArgumentException(s: "node missing");
    }
    // rebuild edge set instead of removing while looping
    Set<String> updatedEdges=new LinkedHashSet<>();
    for(String e:edges){
        int arrowIndex=e.indexOf(str: " -> ");
        String left=e.substring(beginIndex: 0,arrowIndex);
        String right=e.substring(arrowIndex+4);
        if(!left.equals(clean)&&!right.equals(clean)){
            updatedEdges.add(e);
        }
    }
    edges=updatedEdges;
}

// removes a list of nodes
public void removeNodes(String[] nodeArray){
    if(nodeArray==null){
        throw new IllegalArgumentException(s: "null input");
    }
    for(int i=0;i<nodeArray.length;i++){
        removeNode(nodeArray[i]);
    }
}
```

```java
// removes one edge
public void removeEdge(String from,String to){
    if(from==null||to==null){
        throw new IllegalArgumentException(s: "null edge");
    }
    String f=from.trim();
    String t=to.trim();
    if(f.isEmpty()||t.isEmpty()){
        throw new IllegalArgumentException(s: "empty edge");
    }
    String target=f+" -> "+t;
    boolean removed=edges.remove(target);
    if(!removed){
        throw new IllegalArgumentException(s: "edge missing");
    }
}
```

Add test cases for these 3 APIs. The test cases should cover at least the following scenarios:

- Scenario 1: some nodes and some edges are correctly removed.
- Scenario 2: removing nodes that do not exist in the graph will cause exceptions.
- Scenario 3: removing edges that do not exist in the graph will cause exceptions

```
PROBLEMS 2   OUTPUT   TERMINAL   PORTS
TERMINAL
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu:CSE464-2026-frgibbs >--------------------
[INFO] Building CSE464-2026-frgibbs 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE464-2026-frgibbs ---
[INFO] skip non existing resourceDirectory C:\Users\speed\Downloads\CSE464-2026-frgibbs\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE464-2026-frgibbs ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 3 source files with javac [debug target 11] to target\classes
[WARNING] system modules path not set in conjunction with -source 11
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @
 CSE464-2026-frgibbs ---
[INFO] skip non existing resourceDirectory C:\Users\speed\Downloads\CSE464-2026-frgibbs\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE464-2026-frgibbs ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 11] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 11
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ CSE464-2026-frgibbs ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.096 s
[INFO] Finished at: 2026-04-03T15:39:52-07:00
[INFO] ------------------------------------------------------------------------
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs>
```

How to Run

1. The main code is located in: src/main/java/Graph.java
2. The test cases are located in: src/test/java/GraphTest.java
3. Open a terminal in the project folder (where pom.xml is located).
4. Run: mvn package
5. Maven will compile the code and run all test cases.
6. Verify the output shows:
    - BUILD SUCCESS
    - Failures: 0

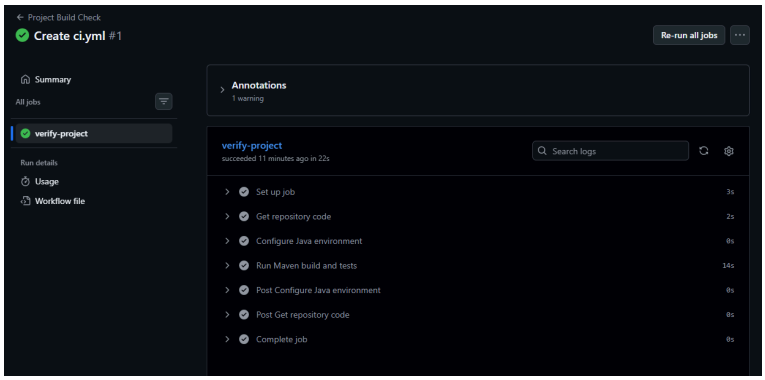
- Errors: 0

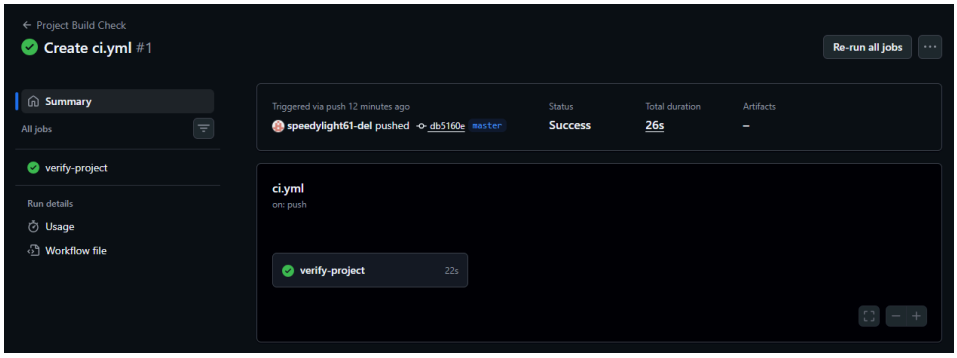
Question 2:

Add continuous integration support for your github project

I set up continuous integration using GitHub Actions for my project. I created a workflow file in the repository that runs automatically whenever code is pushed to the master branch or when a pull request is made. The workflow checks out the code, sets up Java 11, and runs the command “mvn clean package” to build the project and execute all test cases. This ensures that the code compiles correctly and all tests pass every time changes are made.

The workflow ran successfully and showed a green check, which confirms that the build and tests completed without any errors.





How to Run

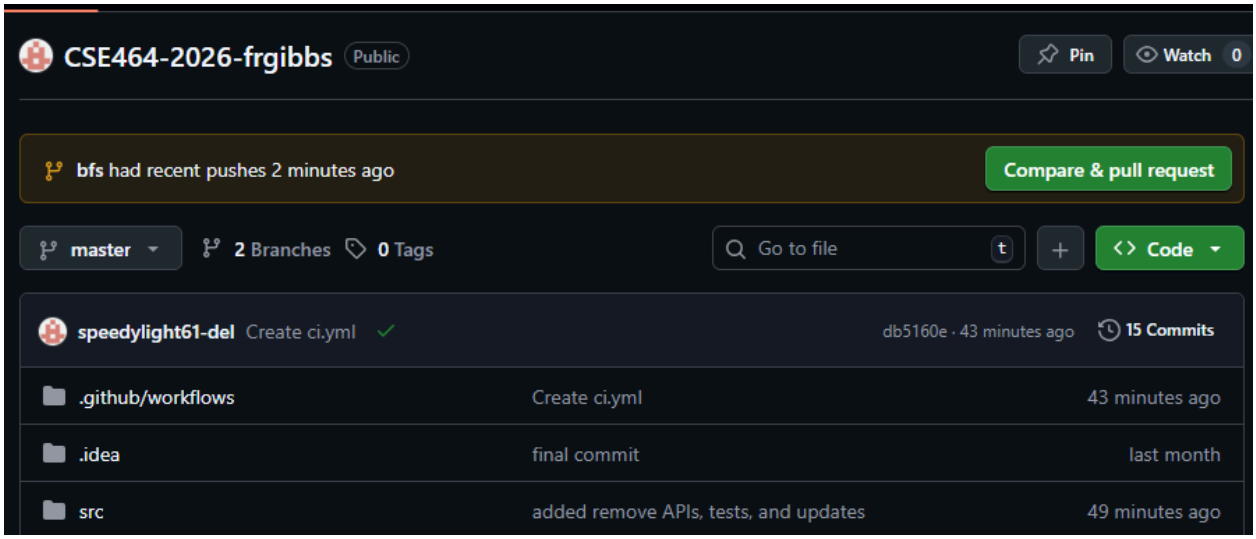
The workflow is located in: .github/workflows/ci.yml

1. Go to the GitHub repository.
2. Navigate to the Actions tab.
3. Select the workflow named “Project Build Check.”
4. Click on the most recent workflow run.
5. View the job verify-project to see the steps executed:
    - Checkout repository

    - Set up Java 11
    - Build and test using Maven
6. Confirm that the workflow shows a green check

Question 3:

Create a branch bfs:



How to Run

1. The BFS implementation is located in: src/main/java/Graph.java
2. The BFS test cases are located in: src/test/java/GraphTest.java
3. Switch to the bfs branch: git checkout bfs
4. Open a terminal in the project folder where pom.xml is located.
5. Run: mvn test
6. Verify the output shows:
    a. BUILD SUCCESS
    b. Failures: 0
    c. Errors: 0

Question 4:

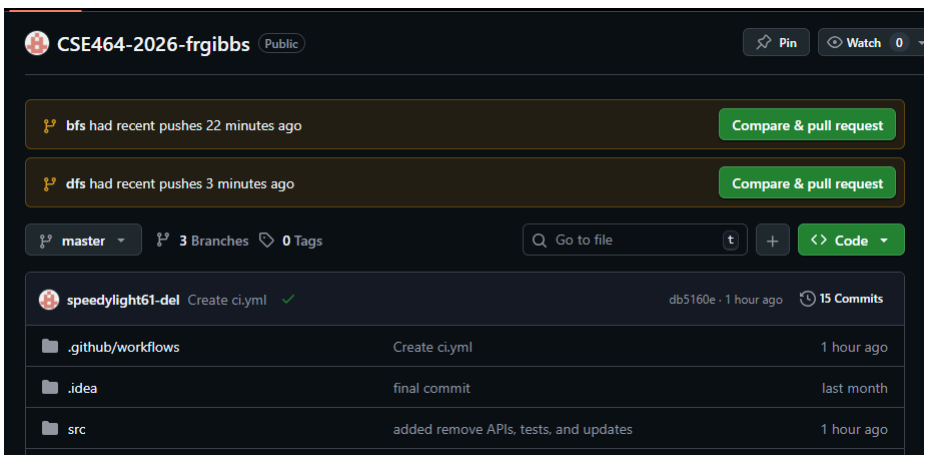
Create another branch dfs, and implement the same graph search API using DFS algorithm

The test cases are located in src/test/java/GraphTest.java. They check that the graph search works correctly by verifying that a valid path is returned when one exists and that null is returned when no path

is found. The same tests are used on both the bfs and dfs branches, so passing results confirm that both search methods work correctly.

```
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu:CSE464-2026-frgibbs >--------------------
[INFO] Building CSE464-2026-frgibbs 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ CSE464-2026-frgibbs ---
[INFO] Deleting C:\Users\speed\Downloads\CSE464-2026-frgibbs\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE464-2026-frgibbs ---
[INFO] skip non existing resourceDirectory C:\Users\speed\Downloads\CSE464-2026-frgibbs\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE464-2026-frgibbs ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 11] to target\classes
[WARNING] system modules path not set in conjunction with -source 11
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @
 CSE464-2026-frgibbs ---
[INFO] skip non existing resourceDirectory C:\Users\speed\Downloads\CSE464-2026-frgibbs\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE464-2026-frgibbs ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 11] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 11
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ CSE464-2026-frgibbs ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running GraphTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.136 s -- in GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.129 s
[INFO] Finished at: 2026-04-03T17:05:06-07:00
```

```java
// Q4: DFS search to find a path between two nodes
public String graphSearch(String start, String end) {
    if (start == null || end == null) return null;
    start = start.trim();
    end = end.trim();
    if (!nodes.contains(start) || !nodes.contains(end)) return null;

    Set<String> visited = new LinkedHashSet<>();
    Map<String, String> parent = new HashMap<>();

    dfsHelper(start, visited, parent);

    if (!visited.contains(end)) return null;

    String path = end;
    String cur = end;

    while (parent.containsKey(cur)) {
        cur = parent.get(cur);
        path = cur + " -> " + path;
    }

    return path;
}

// helper method for DFS
private void dfsHelper(String current, Set<String> visited, Map<String, String> parent) {
    visited.add(current);
    for (String e : edges) {
        int i = e.indexOf(str: " -> ");
        String left = e.substring(beginIndex: 0, i);
        String right = e.substring(i + 4);
        if (left.equals(current) && !visited.contains(right)) {
            parent.put(right, current);
            dfsHelper(right, visited, parent);
        }
    }
}
```



How to Run

1. The DFS implementation is located in: src/main/java/Graph.java
2. The DFS test cases are located in:src/test/java/GraphTest.java
3. Switch to the dfs branch: git checkout dfs
4. Open a terminal in the project folder where pom.xml is located.
5. Run: mvn test
6. Verify the output shows:
    - BUILD SUCCESS
    - Failures: 0
    - Errors: 0

Question 5:

Merge the bfs branch to the main branch, and then merge the dfs branch to the main branch. You should see merge conflict. Resolve the merge conflict by changing the search API to add an enum parameter for choosing a search algorithm.

```
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs> git pull origin m
aster
From https://github.com/speedylight61-del/CSE464-2026-frgibbs
 * branch            master     -> FETCH_HEAD
Updating 74aaf23..db5160e
Fast-forward
 .github/workflows/ci.yml | 26 ++++++++++++++++++++++++
 1 file changed, 26 insertions(+)
 create mode 100644 .github/workflows/ci.yml
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs> git merge bfs
Merge made by the 'ort' strategy.
 src/main/java/Graph.java     | 135 ++++++++++++++++++++++++++++++-
-------------
 src/test/java/GraphTest.java |  17 ++++++
 2 files changed, 110 insertions(+), 42 deletions(-)
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs> git merge dfs
Auto-merging src/main/java/Graph.java
CONFLICT (content): Merge conflict in src/main/java/Graph.java
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs>
```

```
PS C:\Users\speed\Downloads\CSE464-2026-frgibbs> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu:CSE464-2026-frgibbs >--------------------
[INFO] Building CSE464-2026-frgibbs 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ CSE464-2026-frgibbs ---
[INFO] Deleting C:\Users\speed\Downloads\CSE464-2026-frgibbs\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE464-2026-frgibbs ---
[INFO] skip non existing resourceDirectory C:\Users\speed\Downloads\CSE464-2026-frgibbs\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE464-2026-frgibbs ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 11] to target\classes
[WARNING] system modules path not set in conjunction with -source 11
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @
 CSE464-2026-frgibbs ---
[INFO] skip non existing resourceDirectory C:\Users\speed\Downloads\CSE464-2026-frgibbs\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE464-2026-frgibbs ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 11] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 11
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ CSE464-2026-frgibbs ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running GraphTest
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.142 s -- in GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.963 s
[INFO] Finished at: 2026-04-03T18:04:28-07:00
[INFO] ------------------------------------------------------------------------
```

How to Run

1. The merged graph search implementation is located in: src/main/java/Graph.java
2. The test cases are located in: src/test/java/GraphTest.java
3. Switch to the master branch: git checkout master
4. Open a terminal in the project folder where pom.xml is located.
5. Run: mvn test
6. Verify the output shows:
    a. BUILD SUCCESS
    b. Failures: 0
    c. Errors: 0
7. The search method supports both algorithms:
    a. Graph.Algorithm.BFS
    b. Graph.Algorithm.DFS

The test cases verify that both BFS and DFS return a valid path when one exists and return null when no path is found. The tests confirm that both algorithms work correctly after merging.